

Chapter 16

Bargaining

Learning Objectives

- 16.1 Explain the strategic view of bargaining.
- 16.2 Explain the nonstrategic view of bargaining

Chapter 16 – Main Points

- Bargaining can be modeled as either a simultaneous-move or sequential-move game.
- A player can gain an advantage by 1) turning a simultaneous-move game into a sequential move game with a first-mover advantage; or by 2) committing to a position.
- Credible commitments are difficult to make because they require players to commit to a course of action against their self-interest. Thus, the best threat is one you never have to use.
- The **strategic view of bargaining** focuses on how the outcome of bargaining games depends on who moves first and who can commit to a bargaining position, as well as whether the other player can make a counteroffer.
- The **non-strategic view of bargaining** does not focus on the explicit rules of the game to understand the likely outcome of the bargaining. This view focuses on the gains from bargaining relative to alternatives.
- The gains from agreement relative to the alternatives to agreement determine the terms of any agreement.
- Anything you can do to increase your opponent's gains from reaching agreement or to decrease your own will improve your bargaining position.

Warm-Up

- Think of the last time you negotiated something: a price, a deadline, a grade, or even chores. Who had more power, and why? How did that affect the outcome?

Bargaining

- Bargaining comes in two varieties: the strategic view of bargaining and the non-strategic view of bargaining.
- In the strategic view, we set up the bargain as we would a game, either in the simultaneous (normal form) or sequential forms
- In the non-strategic view, we set up the bargain by looking at how we can change a player's bargaining position by altering different circumstances.

Strategic View of Bargaining

- In the strategic view, we most often set up the bargain as a game of chicken.
- Doing so gives a player power to commit and makes the threat more credible
- Consider two scenarios: 1) a teachers' union strikes during the last month of the summer and wants better benefits than they currently receive and 2) a group of workers is trying to form a union at their local shop during a downturn in demand for their products
- Which group is likely to have more credibility and ability to commit to a position?
 - The teachers' union is more likely to commit to its position than the group of workers.

Teachers' Union Game

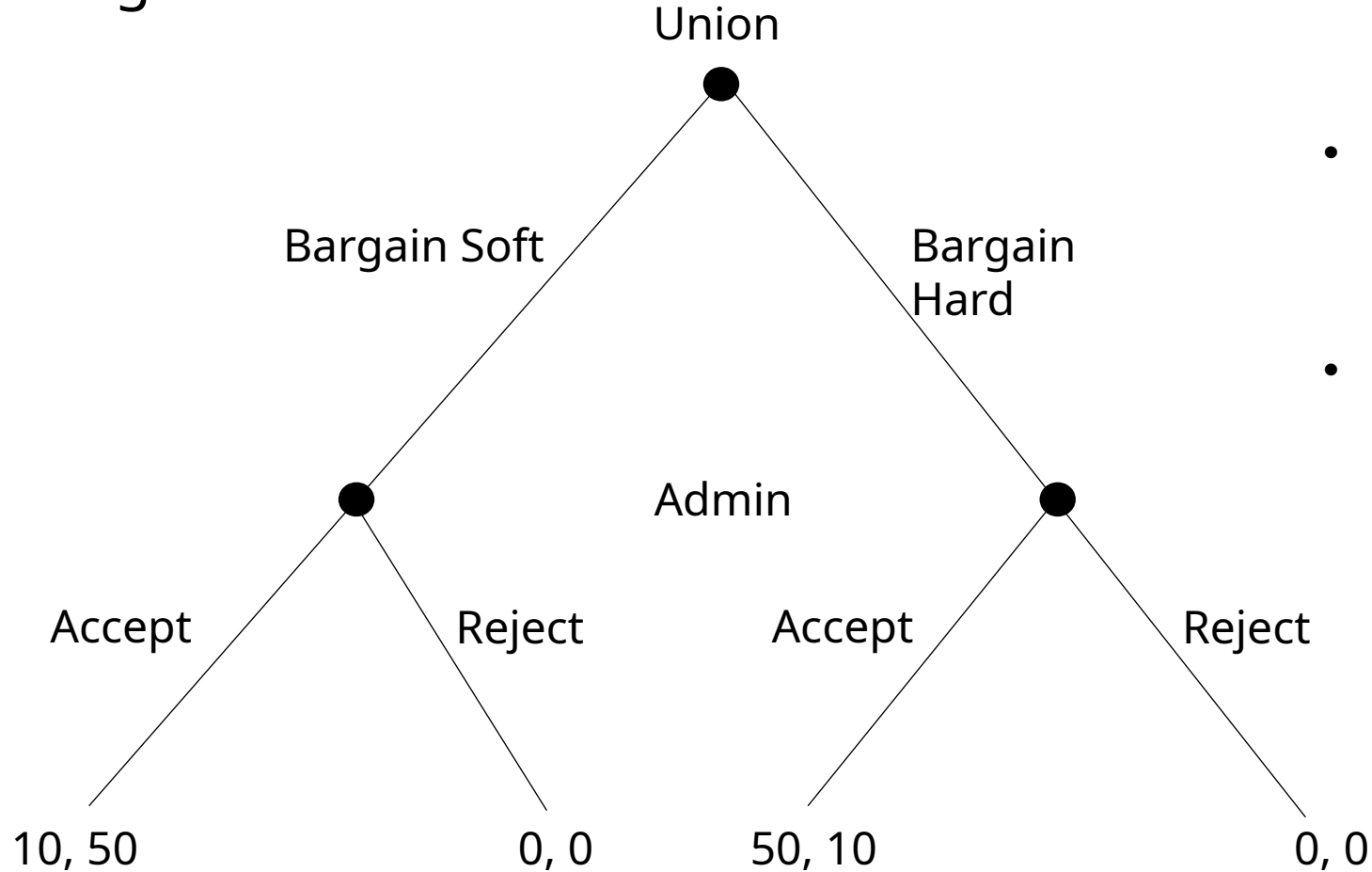
We set up this game of chicken between the teachers' union and the county school administration

		Teachers' Union	
		Bargain Hard	Accommodate
School Admin	Bargain hard	\$0 , \$0	\$50k , \$10k
	Accommodate	\$10k , \$50k	\$30k , \$30k

Two Nash equilibria emerge: Bargain Hard, Bargain Hard and Accommodate, Accommodate

Teachers' Union Game

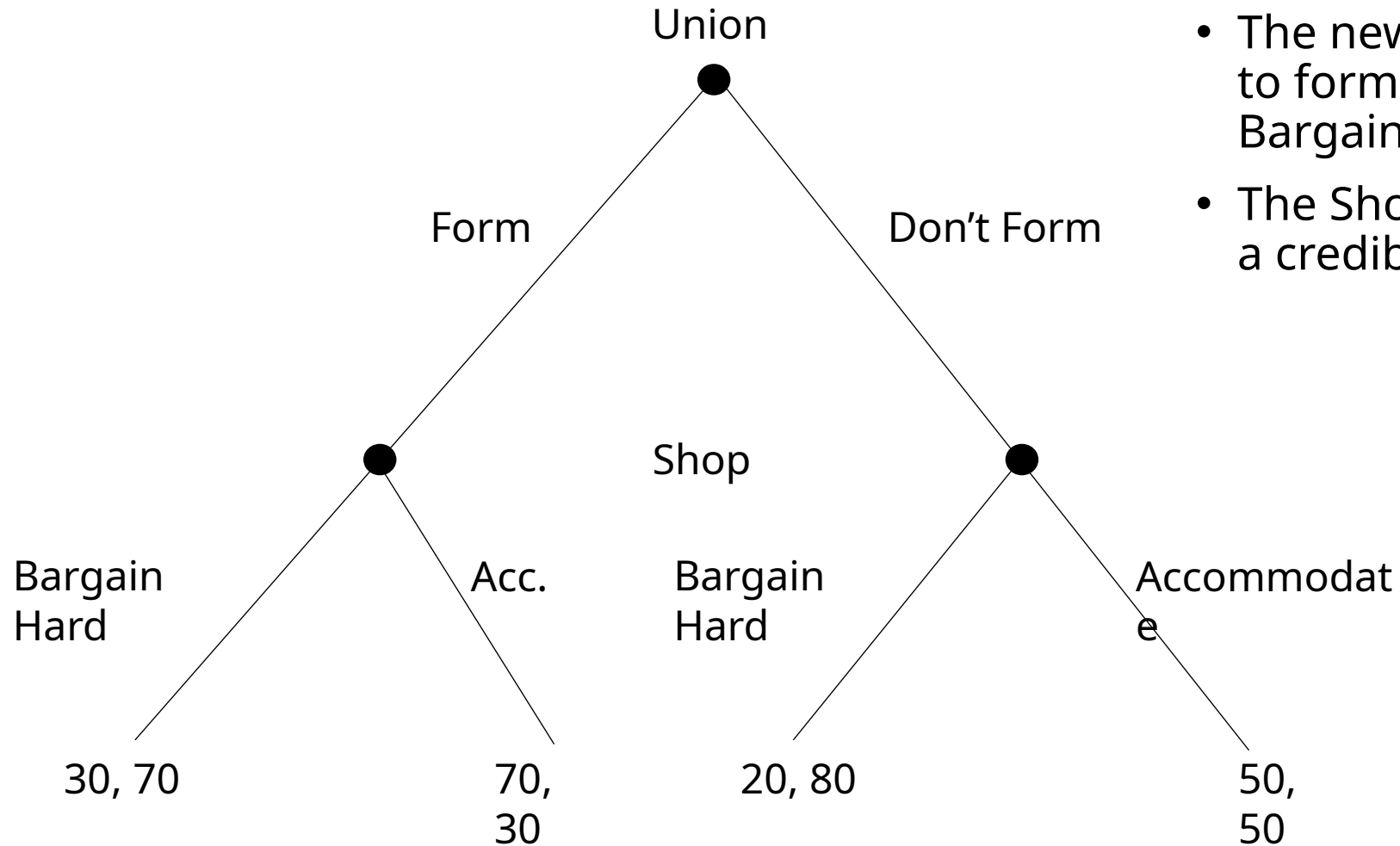
We now characterize the game as a sequential one, where we can see how strong a commitment is.



- If the Union can move first, then they will choose Bargain Hard as their strategy.
- This shows first mover advantage

New Shop Union Game

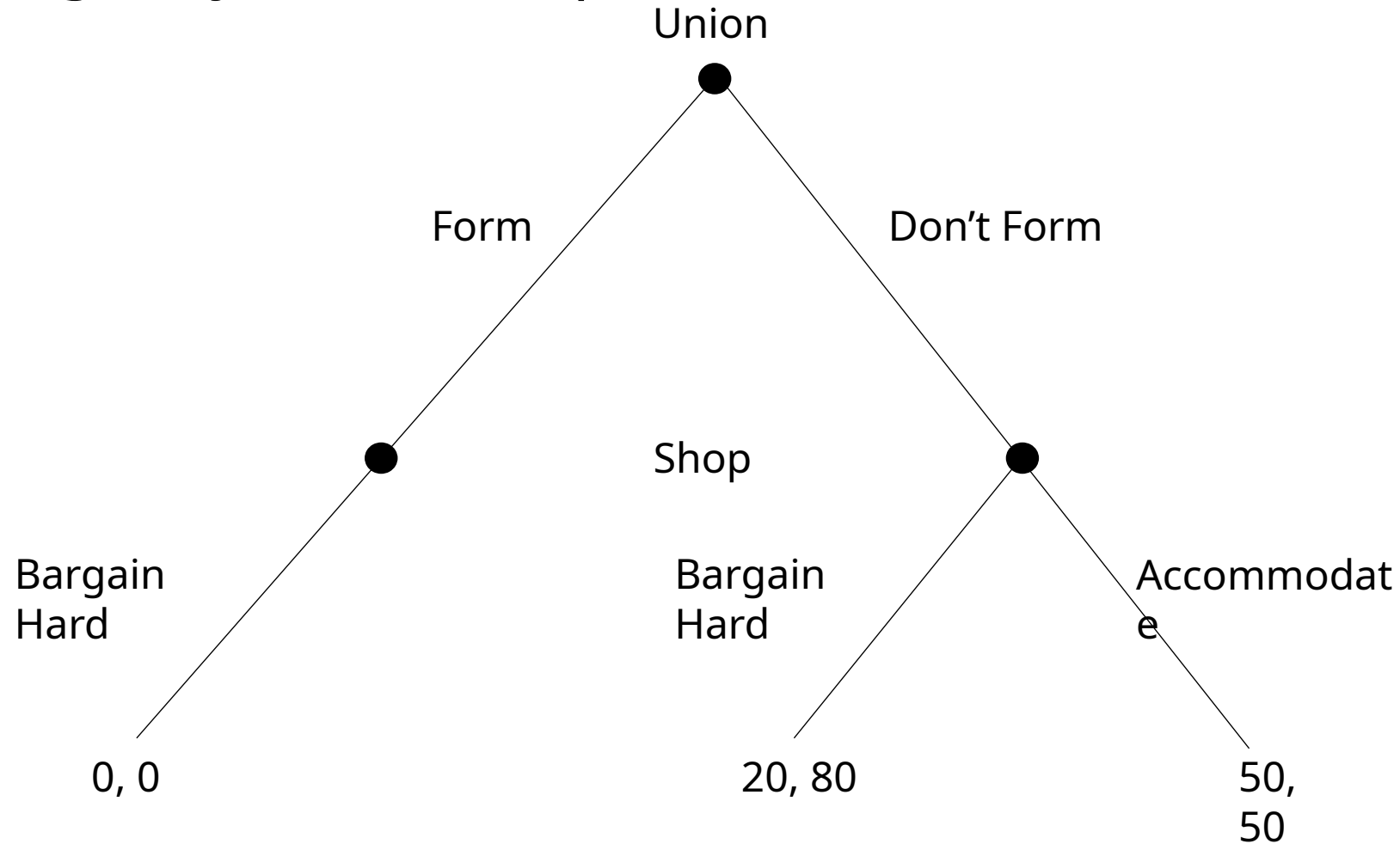
The first-mover advantage in the new shop union game



- The new union will choose to form, but the Shop will Bargain Hard
- The Shop is able to make a credible commitment

New Shop Union Game

Management can commit early and eliminate a strategy by declaring they won't accept unionization



Credible Threats that Aren't so Credible

- In the previous bargains, the party bargaining for a change was potentially faced with a credible threat.
 - Teachers wanted benefits and won't teach until they get them
 - The shop owner outright rejects and unionization of workers
- What if the threat doesn't seem that credible?
 - We hope that our teachers actually care about the kids they are teaching and will relent
 - Can the shop owner afford to lose all his employees at the same time?
- This leads us to the maxim:
 - *The best threat is one you never have to use*
 - We would like to avoid having to find out if the threats are actually true because it will be detrimental to both parties.

Activity – Negotiation Game

Scenario:

- Consider the following game. Two parties are negotiating a wage contract. If they both bargain hard they each gain nothing. If one party bargains hard and the other accommodates, then the bargaining party gets \$10,000 and the accommodating party gets \$2,000. If they both accommodate, each gets \$6,000.

Questions:

1. What is the Nash equilibrium?
2. If the employee threatens to quit, what is the employer's best response?
3. If the employee has already turned in a two weeks notice before negotiations, then what is the firm's best response?

Activity – Negotiation Game

		Employee	
		Bargain Hard	Accommodate
Employer	Bargain hard	0 , 0	\$10k , \$2k
	Accommodate	\$2k , \$10k	\$6k , \$6k

1. What is the Nash equilibrium?
 - (Acc., Bargain Hard) (Bargain Hard, Acc.)
2. If the employee threatens to quit, what is the employer's best response?
 - To accommodate
3. If the employee has already turned in a two weeks notice before negotiations, then what is the firm's best response?
 - To accommodate

Activity – Local Haggling

Scenario:

- Consider two people haggling over a price at the local flea market. The item in question is a piece of junk. The customer has two prices in mind: \$100 (the high price) and \$40 (the low-ball price). The customer values the item at \$120. The seller will either accept the customer's offer or not.

Questions:

1. If the customer offers the high price, what will the seller do?
2. If the customer offers the low price, what will the seller do?
3. Suppose the roles were reversed, what price should the seller charge?
4. What are some credible threats the seller can implement to ensure the high price?

Answers

1. If the customer offers the high price, what will the seller do?
 - Accept the high price and kindly thank them for their business
2. If the customer offers the low price, what will the seller do?
 - Accept the low price
3. Suppose the roles were reversed, what price should the seller charge?
 - The high price since they can extract more producer surplus by doing so.
4. What are some credible threats the seller can implement to ensure the high price?
 - Display “No Haggling” signs in the store, have a reputation for throwing customers out of stores for low-balling them.

Discussion Activity

- Why might a party want to commit to a tough bargaining position, even if it seems risky?
- Think about credibility, first-mover advantage, and signaling intent. When does it give leverage?

Nonstrategic View of Bargaining

- In the strategic view of bargaining, there were well-defined rules that led to possible outcomes in the bargain.
- However, such well-defined rules don't always exist, and we have to have a way to work around the bargain.
- We can do these by using the nonstrategic view of bargaining as "it does not depend on the rules of the bargaining game or whether players can commit to a position."
- For example, UAW is currently negotiating for a better benefits. What provides the GM with any bargaining power?
 - The economy, sales, etc.

Nonstrategic View of Bargaining

- How can we formally view this type of bargaining?
- The key is to understand each players' "outside options" or "disagreement values".
- These are essentially things that increase the bargaining power of one of the players involved.
- For example, if you are negotiating with your boss over a raise, but have no offers from other jobs, you won't have nearly as much bargaining power.
- If your boss has a lot of job applications from potential employees, then his/her disagreement value has increased and yours has decreased in the raise negotiation.

Nonstrategic View of Bargaining

- Formally,
- Let A be the agreement reached. D_1 and D_2 are the disagreement values respectively for players 1 and 2.
- A positive gains from trade in the bargain is defined as:

$$A - (D_1 + D_2) > 0$$

- Under John Nash's axiom, the gains are split evenly such that:

Party 1 gets: $(A + D_1 - D_2)/2$

Party 2 gets: $(A + D_2 - D_1)/2$

Nonstrategic View of Bargaining

- Knowing how to bargain non-strategically is all about changing the disagreement value of the player
- If you want to increase your bargaining power, you need to either
 - 1) increase your disagreement value,
 - 2) decrease the other player's disagreement value or
 - 3) do both
- Having done any of the above and you can improve your bargaining position.
- Essentially, you are making it harder for the other player to bargain with you because you have an improved position that allows you to walk away from the deal.
- This causes the other player to be more accepting of your terms.

Discussion Activity 2

- If you're negotiating a raise, what actions can increase your disagreement value?
- Consider job offers, performance, and the firm's alternatives. What makes your position stronger?

Activity – Used Car Purchase

- Consider purchasing a used car, where haggling is a bit easier.
- You value a particular car at \$15,000 and the dealer values the car at \$10,000—that is the dealer would be willing to sell you the car at \$10,000 but doesn't want to.
- Further, you know there are similar cars at a nearby dealership where you can purchase a car for \$14,000 (scenario 1), and the dealer can sell the same car to another buyer for \$16,000 (scenario 2).

Activity – Used Car Purchase

- Consumer's value = \$15,000
 - Dealer's minimum acceptable price = \$10,000
 - Maximum consumer surplus = $\$15,000 - \$10,000 = \$5,000$
 - Maximum willingness to pay = \$14,000 (Scenario 1)
 - Potential joint gains from trade = \$4,000 (from \$10,000 to \$14,000)
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- Dealer's alternative to sell at \$16,000 (Scenario 2)
 - Consumer won't pay more than \$15,000
 - No deal will be made and no potential gains from trade

Activity – Used Car Purchase

- Disagreement value: the value you get from buying a similar car elsewhere
- You: $\$15,000 - \$14,000 = \$1,000$
- The \$1,000 represents what you can fall back to (the first deal).
 - You can tell the first dealer you won't pay more than \$14,000 since you have outside options.

Scenario 2:

- You: still \$1,000
- Dealer: $\$16,000 - \$10,000 = \$6,000$
- The dealer has increased their outside options. They can now sell to another customer with a higher willingness to pay.

Activity – Used Car Purchase

Scenario 1: you purchase the car for \$14,000.

- Total surplus = $\$15,000 - \$10,000 = \$5,000$
- You get \$1,000 ($\$15,000 - \$14,000$)
- The dealer gets \$4,000 ($\$14,000 - \$10,000$)
- Not a Nash bargaining solution, since not split evenly
- Change the price to \$12,500. Each party gets \$2,500 in surplus.

Scenario 2: No deal

- The dealer has outside options that are better than any deal with you.

Takeaways

- The dealer had a much stronger position in the bargain, as the dealer had the higher opportunity cost of selling the car
- The buyer had a lower opportunity cost and thus a lower disagreement value
- Had the dealer had no other potential buyers, then his disagreement value would have decreased, and his bargaining power lowered.
- Had the buyer not had an alternative, then his disagreement value would have decreased along with his bargaining power.
- Altering your disagreement value is the key to successfully getting a bargain to work in your favor.

Activity – Disagreement Values

1. Nick is in the market for a car. He knows that car dealers get their commissions at the end of the month. When should Nick go to purchase a car?
2. Nick is in the market for a car. He and a car dealer are haggling over the price of the car. What scenarios will raise Nick's disagreement value?
3. Nick is in the market for a car. He and a car dealer are haggling over the price of the car. What scenarios will raise the dealer's disagreement value?
4. Amy wants a raise. She begins to bargain with her boss. What scenarios will raise her disagreement value?

Activity – Disagreement Values

1. At the end of the month when the car dealer is more willing to lower the price of the car in order to make a sale and earn a commission. Ideally, at the end of the day.
2. If there are no other potential buyers in the store. A potential customer was in talks with the dealer then leaves. If Nick walks in with a cashier's check that can't be changed.
3. Another potential customer walks in the store. He doesn't have a cashier's check on hand. There aren't many or any close substitutes for the car.
4. She has another job offer. She is the most productive member of the staff. She is working on an integral long-term project.

Nonstrategic View and Bonuses

- Bonuses operate to weaken or decrease a player's disagreement value
- If a bonus is offered to the salesman, then the salesman is more likely to try and make a sale even at a lower price
- While more sales may be generated, this has the adverse effect of misaligning the goals of the company with the goals of the salesman (the principal-agent problem)
- To realign the goals of the company and the salesman we will need to consider some additional criteria covered later.

Nonstrategic View Examples

- Seasonal variations are times of the year when a firm's sales are either high or low. Think about sales at Christmas or Black Friday
- Firms that are susceptible to these types of variations in sales also see a weakening of their bargaining power during this time (the opposite for the union or other agent)
- If a firm has unionized employees and they threaten to strike during a time of peak demand, then the firm may have no choice but to accept the terms in the bargain
- If, on the other hand, the firm does the same, but in times of low demand, the union may have no choice but to accept the bargain
- Lastly, consider a firm that can forgo a bargain simply because the cost of the bargain is greater than the loss of the non-agreement. In this case, the firm has more bargaining power than the agent/union.

Strategic vs. Nonstrategic Bargaining Outcomes

Scenario	With Strategic Moves	Without Strategic Moves
Labor Negotiation	Union strikes first, gains leverage	Both sides wait, agreement delays, value erodes
Licensing Deal	Firm pre-commits to exclusivity → better terms	Each side delays, uncertainty lowers deal value
M&A Negotiation	Buyer makes early offer with exit penalty clause	Negotiations drag, competing bidders emerge
Supply Contract	Manufacturer invests in specialized capacity	Buyer doubts long-term commitment, reduces offer

Discussion Activity 3

- Why might a union strike around Black Friday or a company offer raises during a downturn?
- Think about bargaining power shifts from seasonal sales, demand cycles, and opportunity costs.

Case Study: CVS v. Walgreens –Bargaining with Pharmacy Benefit Managers (PBMs)

- Pharmacy Benefit Managers (PBMs) like Express Scripts, OptumRx, and CVS Caremark manage prescription benefits for insurance companies.
- Retail pharmacies (e.g., Walgreens, CVS) must negotiate with PBMs to be included in their prescription networks.
- PBMs want lower reimbursement rates; pharmacies want volume and margin.

Event: Walgreens–Express Scripts Dispute (2011–2012)

- Walgreens and Express Scripts failed to reach a contract renewal agreement.
- Walgreens walked away and refused to accept lower payment terms.
- Express Scripts removed Walgreens from its network.
- Over 50 million members lost Walgreens as a pharmacy option.
- Walgreens lost market share to CVS and smaller chains.
- After a year, Walgreens returned to the table and accepted lower terms.

Case Study: CVS v. Walgreens –Bargaining with Pharmacy Benefit Managers (PBMs)

Strategic View

- Walgreens had a first-mover threat: walk away and test PBM flexibility.
- Express Scripts called the bluff.
- In a sequential game, Express Scripts had stronger outside options (other pharmacy partners).

Nonstrategic View

- Walgreens' outside option was weak: lost sales and foot traffic.
- Express Scripts had many alternative partners, weakening Walgreens' bargaining position.

Lessons

- Strategic threat (walking away) must be credible and sustainable.
- Disagreement values shape leverage: Walgreens' value outside the deal was too low.
- Timing matters. Prolonged exclusion hurt Walgreens more than Express Scripts.
- The blend of strategic commitment and economic fallback defines real-world bargaining power.

Key Takeaways

- There are two main views of bargaining:
 - The strategic view focuses on how parties interact like players in a game (e.g., threats, timing, signaling).
 - The nonstrategic view simplifies the process by dividing gains based on outside options.
- The side with better alternatives (a stronger "disagreement value") tends to secure more favorable outcomes.
- Disagreement values define your leverage. Raising your own fallback option or lowering the other party's makes your position stronger.
- Timing and commitment matter. First movers, deadlines, and credible threats can dramatically shape negotiation results.
- Even cooperation requires strategy. Success in bargaining is about pushing harder, understanding incentives, alternatives, and credible positioning.

Knowledge Check

1. What are the two main views of bargaining discussed in this chapter?
2. What is a “first-mover advantage” in bargaining?
3. What are disagreement values and how do they affect outcomes?
4. When is a threat considered credible?
5. How can bonuses weaken a salesperson’s disagreement value?

Knowledge Check Answers

1. Strategic view (modeled like a game) and nonstrategic view (focuses on alternatives/disagreement values).
2. The ability to influence outcomes by committing to a strategy before the other party moves.
3. The value of an agent's outside options; higher values improve bargaining position.
4. A threat is credible if it's believable, observable, and potentially costly to ignore.
5. They increase motivation to close a deal, making it harder for the salesperson to walk away.

Consultant's Challenge: Structuring the Bargain

Scenario:

- You've been hired by a labor union to prepare for a high-stakes negotiation with a major manufacturer. The union wants better health benefits and pay raises, but the employer is threatening layoffs if costs rise.

Your Task:

1. How would you analyze this negotiation using both the strategic and nonstrategic views of bargaining?
2. What actions would you recommend to strengthen the union's bargaining position before formal talks begin?
3. What are some credible signals or commitments the union might make to improve outcomes?